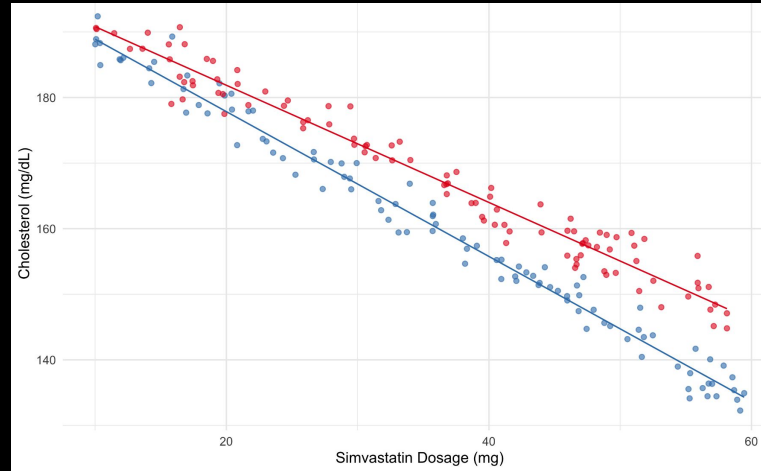


Generalized Linear Models - II and Review



Saket Choudhary

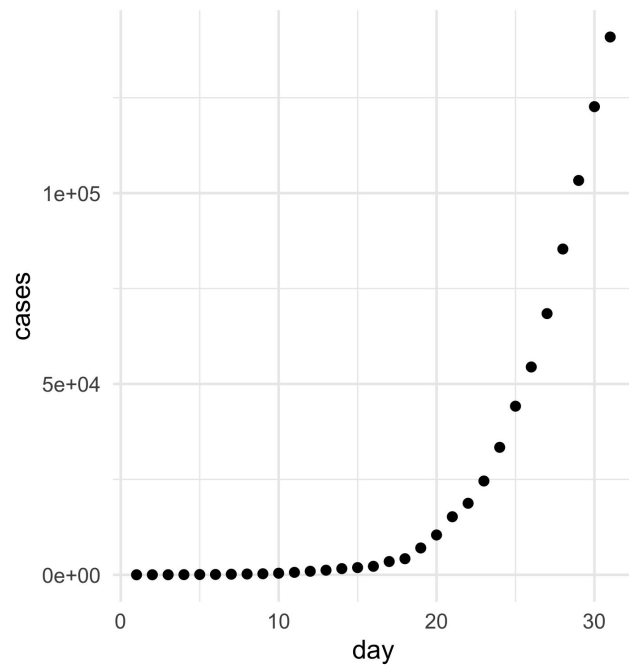
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Introduction to Public Health Informatics

DH 302

Lecture 14 || Friday, 21st February 2025

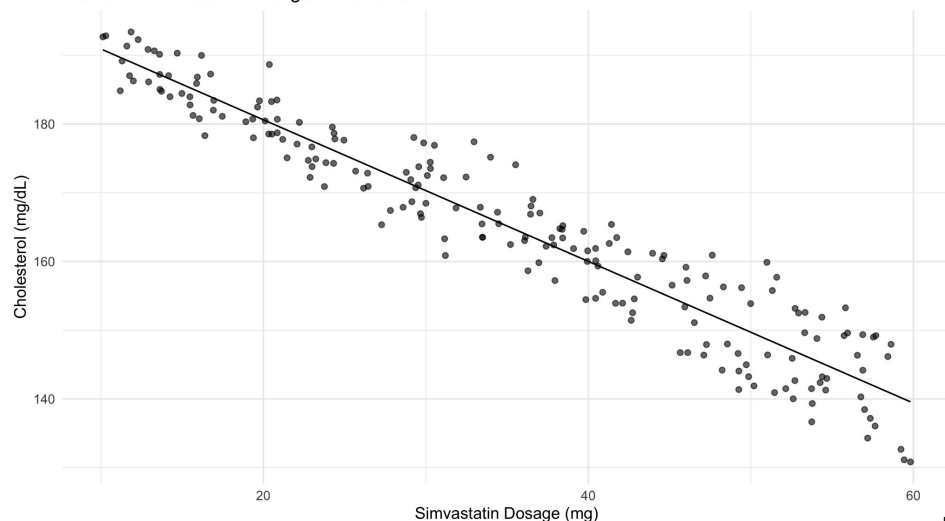
Covid Example



A simple linear regression problem: Effect of drug on cholesterol

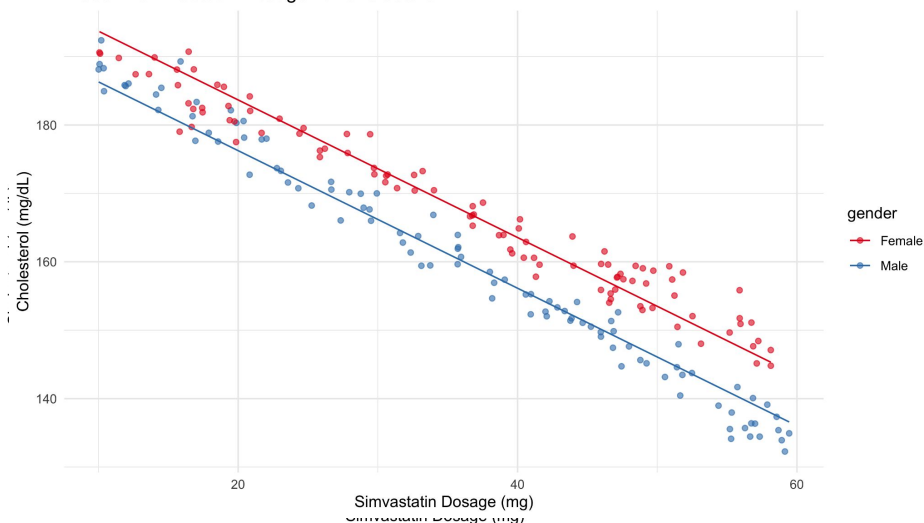
All values hypothetical

Model1: Simvastatin Dosage vs. Cholesterol



$\text{cholesterol} = b_0 + b_1 \cdot \text{dosage}$
 b_0 = cholesterol when no drug (overall intercept)
 b_1 = mean change in cholesterol per unit change in dosage

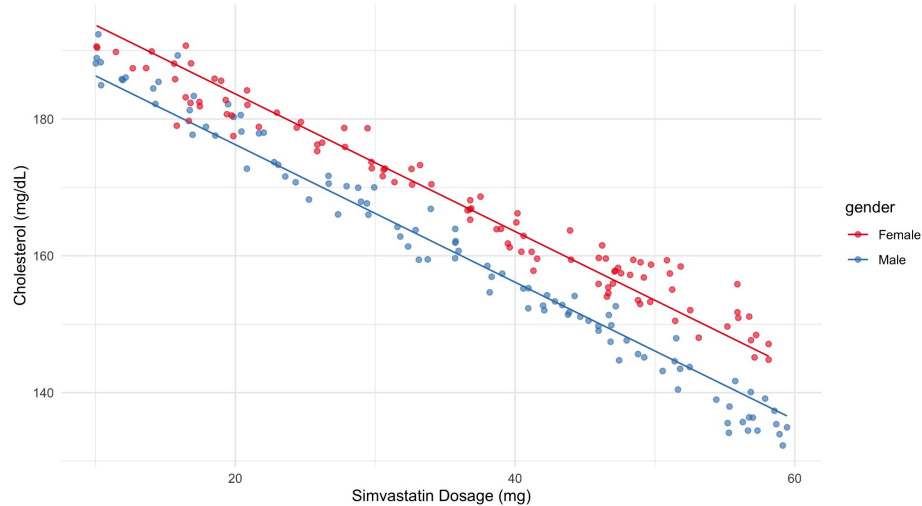
Model2: Simvastatin Dosage vs. Cholesterol



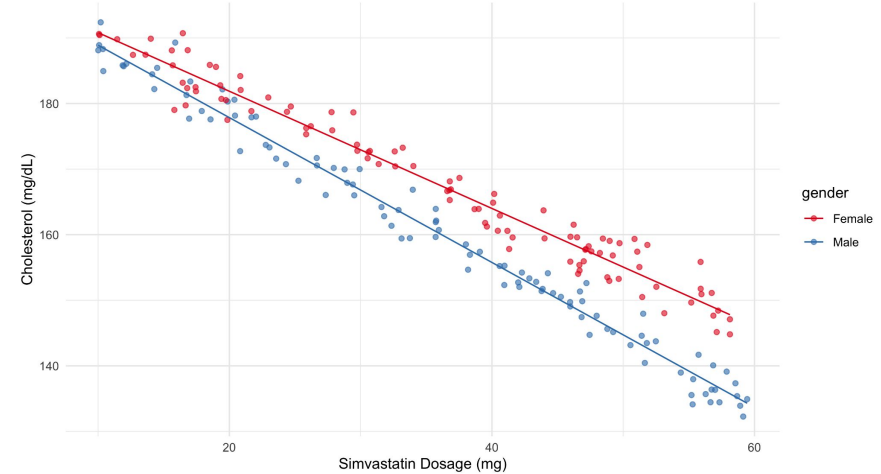
$\text{cholesterol} = b_0 + b_1 \cdot \text{dosage} + b_2 \cdot \text{gender}$
 b_0 = cholesterol when no drug for females (intercept of females)
 b_1 = mean change in cholesterol per unit change in dosage
 b_2 = Mean change in cholesterol for males (wrt females) when dosage is constant (intercept shift for males)
Male intercept = $b_2 + b_0$; Male slope = b_1
Female intercept = b_0 ; Female slope = b_1

Interaction effects

Model2: Simvastatin Dosage vs. Cholesterol



Model3: Simvastatin Dosage vs. Cholesterol



$\text{cholesterol} = b_0 + b_1 \cdot \text{dosage} + b_2 \cdot \text{gender}$
 b_0 = cholesterol when no drug for females (intercept of females)
 b_1 = mean change in cholesterol per unit change in dosage
 b_2 = Mean change in cholesterol for males when dosage is constant (intercept difference for males)
Male intercept = $b_2 + b_0$; Male slope = b_1
Female intercept = b_0 ; Female slope = b_1

$\text{cholesterol} = b_0 + b_1 \cdot \text{dosage} + b_2 \cdot \text{gender} + b_3 \cdot \text{gender} \cdot \text{dosage}$
 b_0 = cholesterol when no drug for females (intercept of females)
 b_1 = mean change in cholesterol per unit change in dosage for females (slope for females)
 b_2 = Mean change in cholesterol for males when dosage is constant (intercept difference for males)
 b_3 = Difference in slope of male wrt females
Male intercept = $b_2 + b_0$; Male slope = $b_1 + b_3$
Female intercept = b_0 ; Female slope = b_1

Crabshoe example: Stepwise variable selection

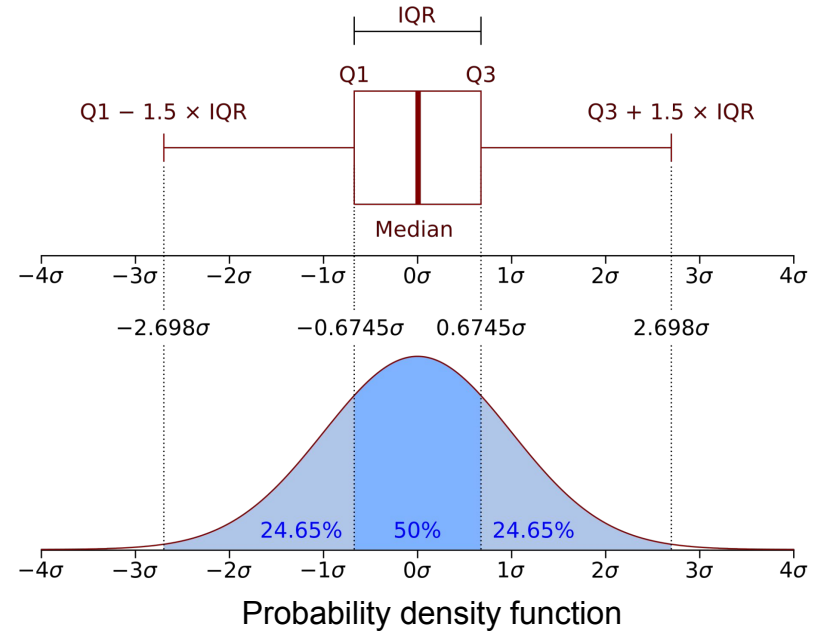
- When doing a multiple regression (LM/GLM), we can use algorithms to select explanatory variables for a model, or delete them, in a stepwise manner – Backward or Forward selection
- But should be used cautiously - the output might not yield a meaningful model
- You should include a variable if it is important for a study even if it turns out to be not statistically significant

Mini Review

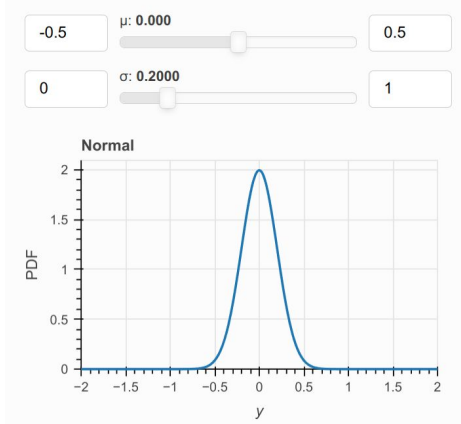
Properties of a PMF/PDF

Suppose we have a process that outputs a discrete realization X that takes exactly J values in a set $X = \{x_1, x_2, \dots, x_J\}$. Assume x_j occurs with probability p_j , $j = 1, 2, \dots, J$, then $p_1, p_2, \dots, p_J$ is called the probability mass (distribution of X is continuous) function as long as it satisfies:

$p_1, p_2, \dots, p_J \geq 0$
 $p_1 + p_2 + \dots + p_J = 1$ Law of total probability



Gaussian a.k.a Normal distribution



Normal

$$f(y; \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(y-\mu)^2/2\sigma^2}.$$

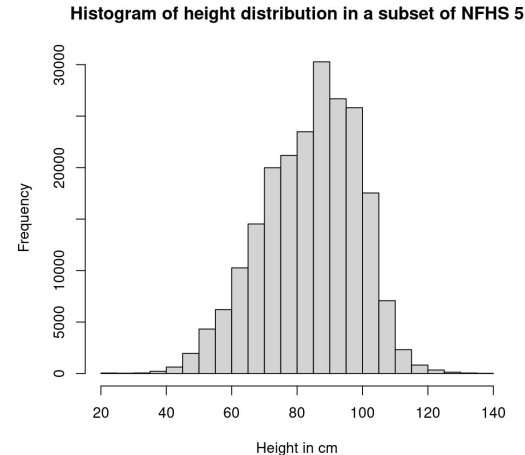
Mean: μ

Variance: σ^2

$$X_i \sim \mathcal{N}(\mu, \sigma^2)$$

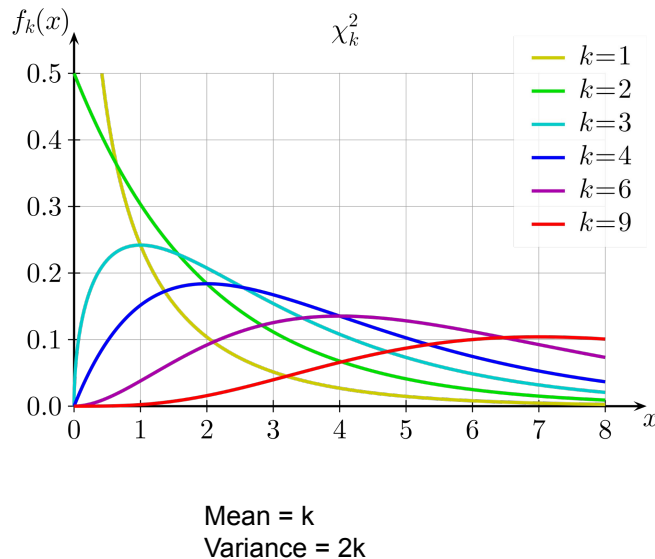
Quantities that are sum of large number of subprocesses tend to be normally distributed

Example: Height/blood pressure distribution of a sample



Chi-square distribution

Given $Z_1, Z_2, \dots, Z_k$ are independent standard normal distribution, i.e. $Z_i \sim \mathcal{N}(0, 1)$, then the sum of their squares follows a χ^2 distribution with k degrees of freedom



$$X = \sum_{i=1}^k Z_i^2$$
$$f(x; k) = \begin{cases} \frac{x^{k/2-1} e^{-x/2}}{2^{k/2} \Gamma\left(\frac{k}{2}\right)}, & x > 0; \\ 0, & \text{otherwise.} \end{cases}$$

$$X \sim \chi_k^2$$
$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt, \quad \Re(z) > 0.$$

Example: Estimate the parameters by curve fitting and check how “good” does it explain the observations

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$$

O_i = an observed count for bin i

E_i = an expected count for bin i

T-test primer

$X_1, \dots, X_n$ are independent realizations of the normally-distributed random variable X

Sample mean $\bar{X}_n = \frac{1}{n}(X_1 + \dots + X_n)$

Sample variance $s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$

T follows student t-distribution $T \equiv \left(\bar{X}_n - \mu \right) \frac{\sqrt{n}}{s}$

Hypothesis testing

- In hypothesis testing, we investigate if the observations could have come by chance or if random processes are sufficient to explain the observations.
- Null hypothesis H_0 : Hypothesis that chance alone is responsible for explaining the observations
- Example: The observations in Mar 2019 - Mar 2023 can be explained by the observations in Feb 2015 - Feb 2019 alone.
- Requires constructing a **statistical model** of what the observations would like if chance or random process can alone explain it
- A “**test-statistic**” measures the deviation of observations from the expectations (i.e. the null hypothesis)
- We measure if the test-statistic deviates from an pre-decided threshold. If it does → We “reject” the null hypothesis, if it does not we “fail to reject” the null hypothesis

Hypothesis testing

Question 2

10 points

Postoperative ileus (POI) is a form of gastrointestinal dysfunction that commonly occurs after abdominal surgery and results in absent or delayed gastrointestinal motility. Does rocking in a chair after abdominal surgery reduce postoperative ileus (POI) duration? Sixty-six postoperative abdominal surgery patients were randomly divided into two groups. The experimental group ($n = 34$) received standard care plus the use of a rocking chair while the control group ($n = 32$) received only standard care. For each patient, the postoperative time until first flatus (days) (an indication that the POI has ended) was measured. The results are tabulated below

	n	Mean (days)	SD
Control	32	3.88	0.80
Rocking	34	3.16	0.86

(a) [5 points] State the null and alternate hypothesis.

Solution: $H_0 : \mu_{\text{control}} = \mu_{\text{rocking}}$ – 1 point

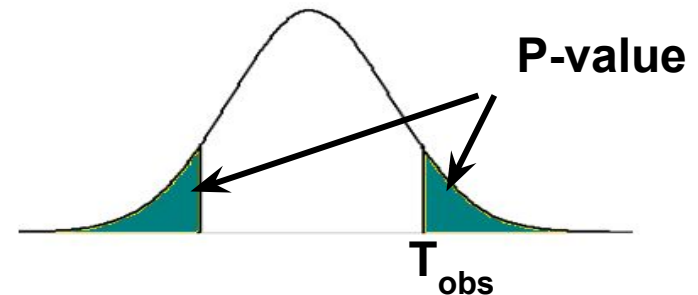
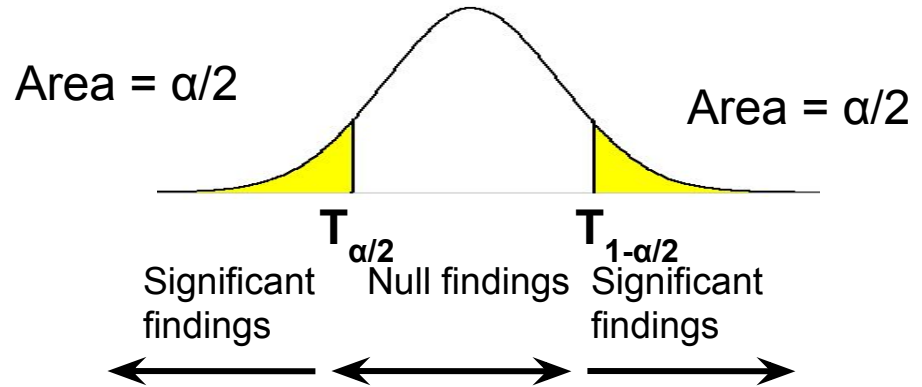
$H_{\text{alt}} : \mu_{\text{control}} > \mu_{\text{rocking}}$ – 4 points

Visualizing the p-values region

P-value = Probability of sampling a test statistic at least as extreme as the observed test statistic if the null hypothesis is true

We “reject” the null hypothesis (H_0) if the pvalue is below the threshold (α)

Distribution of T under H_0



Type I,II errors and Power

- **Type I error:**

- Probability that the test incorrectly rejects the null hypothesis (H_0) when the null H_0 is true
- Often denoted by α

- **Type II error:**

- Probability that the test incorrectly fails to reject the null hypothesis (H_0) when H_0 is false
- Often denoted by β

- **Power:**

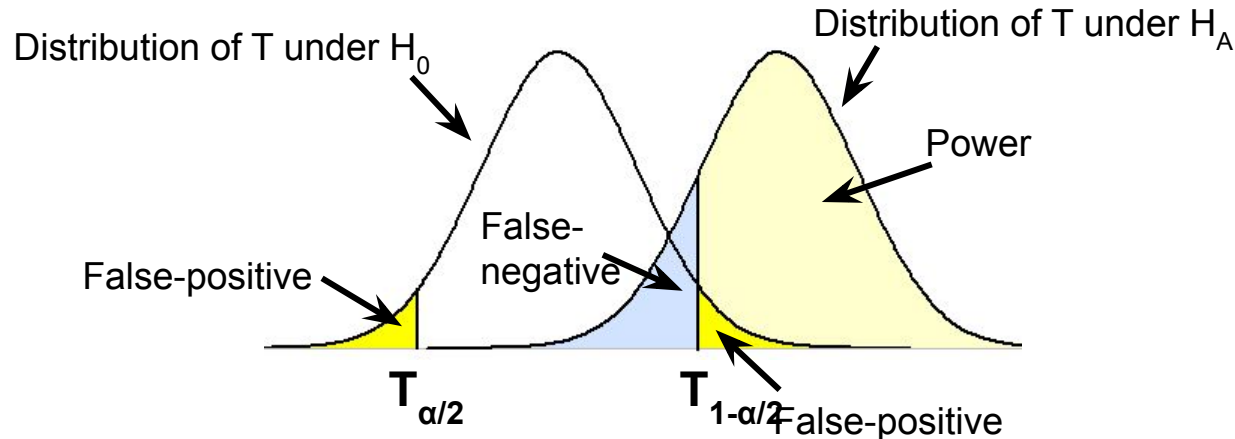
- Probability that the test correctly rejects the null hypothesis (H_0) when the alternative hypothesis (H_1) is true
- Commonly denoted by $1 - \beta$ where β is the probability of making a Type II error by incorrectly failing to reject the null hypothesis.
- As β increases, the power of a test decreases.

Type I,II errors and Power

The **false-positive** rate is the probability of **incorrectly *rejecting*** H_0 .

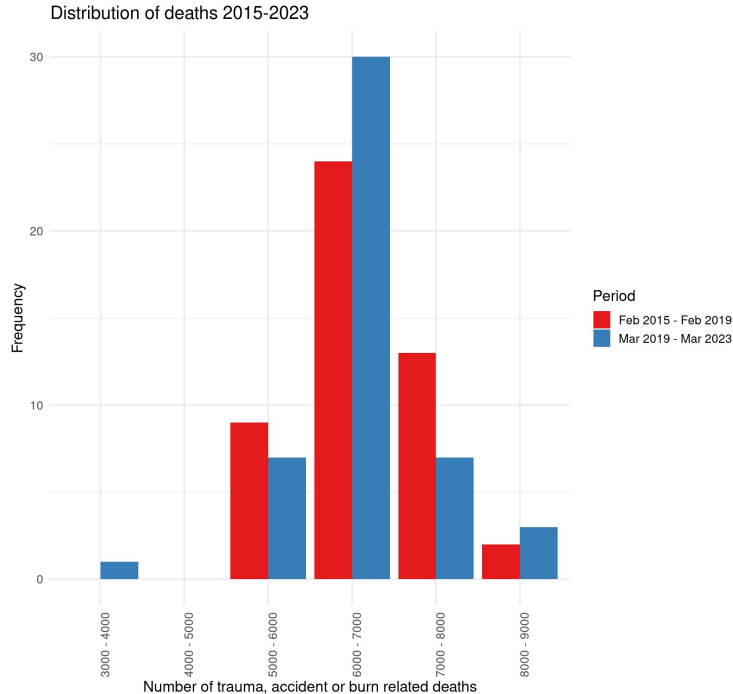
The **false-negative** rate is the probability of **incorrectly *accepting*** H_0 .

Power = $1 - \text{false-negative rate}$ = probability of **correctly rejecting** H_0 .



Goodness of fit - Chi-squared test

Problem: What distribution should I fit?



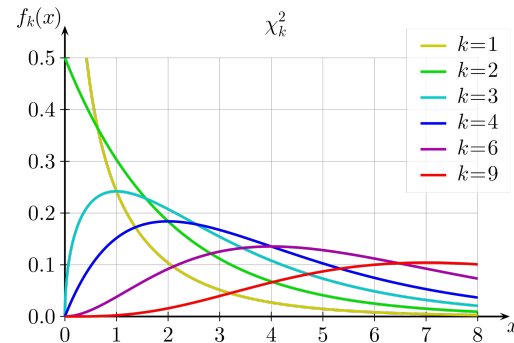
Solution: Quantify how “good” does the expected model (frequencies) explain the observations

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$$

O_i = an observed count for bin i

E_i = an expected count for bin i , asserted by the null hypothesis

Calculate
p-value



Expectations and Variances

The expected value of a random variable is the average value of all expected outcomes. More formally,

$$\begin{aligned}\mathbb{E}[X] &= \sum_{j=1}^J x_j P(X = x_j) \\ &= x_1 p_1 + x_2 p_2 + \cdots + x_J p_J.\end{aligned}$$

The expectation of the sum = Sum of the expectations. More formally,

$$\mathbb{E}[X_1 + X_2 + \cdots + X_N] = \mathbb{E}[X_1] + \mathbb{E}[X_2] + \cdots + \mathbb{E}[X_N] \quad (1)$$

Expectations and Variances

When we observe (realize) different values of $\mathcal{X}$, one natural question to ask is how scattered these values tend to be from the expected value. This is quantified by "variance". More formally,

$$Var[X] = \mathbb{E}[(X - \mathbb{E}[X])^2] \quad (2)$$

$$= \mathbb{E}[X^2 - 2X\mathbb{E}[X] + (\mathbb{E}[X])^2] \quad (3)$$

$$= \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \quad (4)$$

The positive square root of $Var[X]$ is called standard deviation of X .

For independent variables, variation of their sum is equal to sum of individual variances, i.e. when X_i are independent,

$$Var(X_1 + X_2 + \dots X_N) = Var(X_1) + Var(X_2) + \dots Var(X_N)$$

Standard error of the mean

Standard error of the mean:

- $\bar{x}$ is an **estimate of the population mean μ** , but how good of an estimate is it?
- How “good” (close to the population mean) the sample mean $\bar{x}$ It depends on the sample and is often referred to as the sampling error.
- If you could sample the entire population this sampling error will be zero.
- The standard deviation of the sampling distribution of the sample mean is given by: $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$ where n is the number of times sample mean was drawn
- But, we do not always have σ available, so we use its estimate, the sample standard deviation s . The standard error of the mean is then defined as: $SE_{\bar{X}} = \frac{s}{\sqrt{n}}$

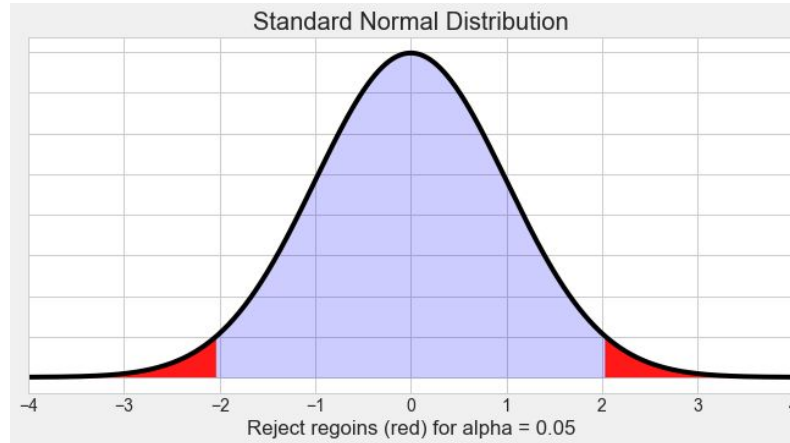
Standard Error vs Standard Deviation:

- Standard deviation (SD) **describes the dispersion of the data**
- **Standard error (SE) describe the unreliability of the mean of the sample** due to sampling error
- As the sample size increases, sample mean and the standard deviation both approach the population mean and standard deviation respectively
- The standard error tends to decrease as the sample size n increases $\rightarrow$ sample mean becomes more and more precise estimate of the population mean.

Testing for the mean of a distribution: Z-test

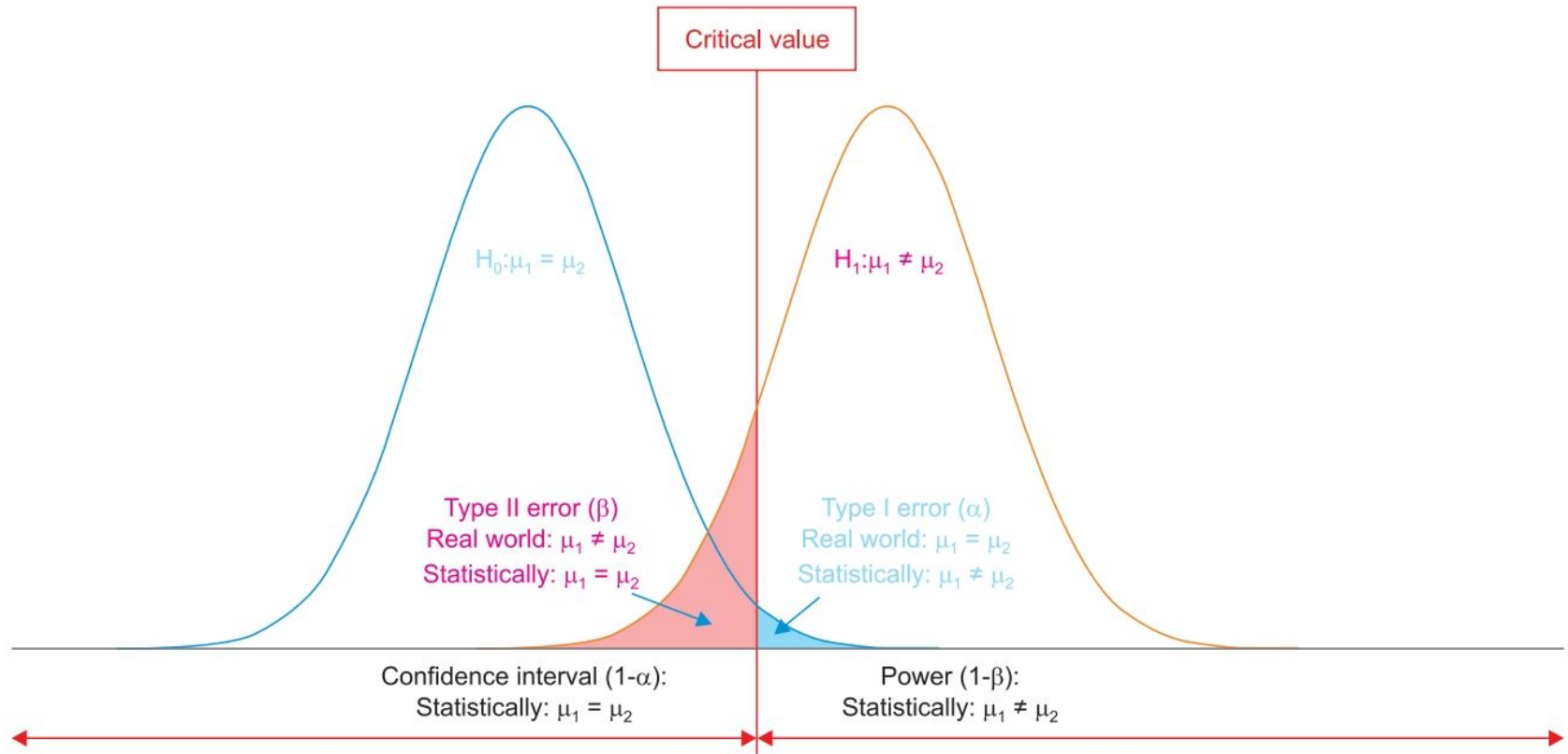
Null hypothesis: Is the mean value of the sample a given quantity

$$Z = \frac{(\bar{X} - \mu_0)}{SE}$$



Only valid if the population variance is known

Testing for difference in mean (median) of two samples



T-test paradigm

$$H_0 : \mu_1 = \mu_2$$

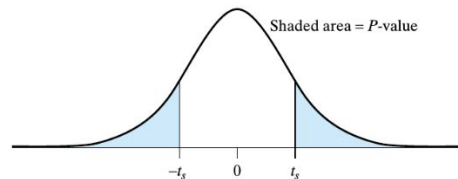
The t-statistic t_s measures how far are the mean difference of the groups $\bar{y}_1 - \bar{y}_2$ from the expectation of zero, if the null were true:

$$t_s = \frac{\bar{y}_1 - \bar{y}_2}{SE_{\bar{y}_1 - \bar{y}_2}}$$

The 's' in t_s indicates that this t-statistic is calculated from the samples. If H_0 is true, then t_s follows a Student's t-distribution approximately with degrees of freedom given by

$$df = \frac{(SE_1^2 + SE_2^2)^2}{\frac{SE_1^4}{n_1 - 1} + \frac{SE_2^4}{n_2 - 1}}$$

t_s Follows a t-distribution with df degrees of freedom



T-test assumptions

- Observations from both the groups are approximately normally distributed
- The difference of means $\bar{y}_1 - \bar{y}_2$ is independent of the standard error of the difference $SE_{\bar{y}_1 - \bar{y}_2}$

T-test: The scenarios

Equal sizes, same variance

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p \sqrt{\frac{2}{n}}};$$

$$s_p = \sqrt{\frac{s_{X_1}^2 + s_{X_2}^2}{2}}$$

$$\text{d.f.} = 2n-2$$

Equal or unequal sizes,
similar variance

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p \cdot \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}},$$

$$\frac{1}{2} < \frac{s_{X_1}}{s_{X_2}} < 2$$

$$s_p = \sqrt{\frac{(n_1 - 1)s_{X_1}^2 + (n_2 - 1)s_{X_2}^2}{n_1 + n_2 - 2}}$$

$$\text{d.f.} = n_1 + n_2 - 2$$

Equal or unequal sizes,
unequal variance (Welch
test)

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_{\bar{\Delta}}}$$

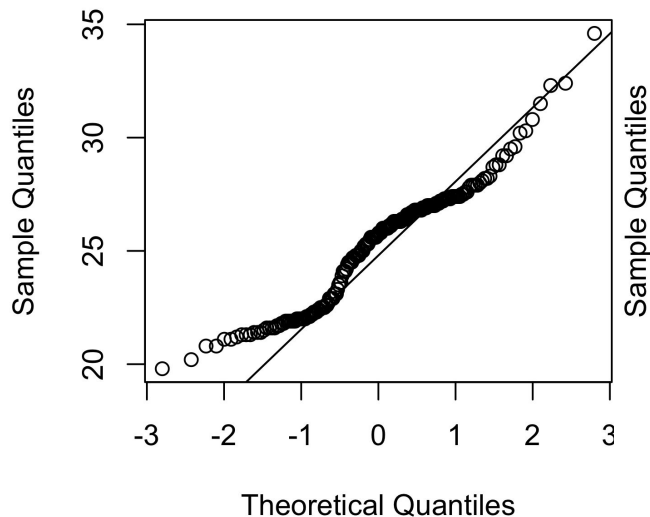
$$s_{\bar{\Delta}} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}.$$

$$\text{d.f.} = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}.$$

Verifying the ‘normality’ assumption using QQplot

```
qqnorm(men_rural_2017$bmi)  
qqline(men_rural_2017$bmi)
```

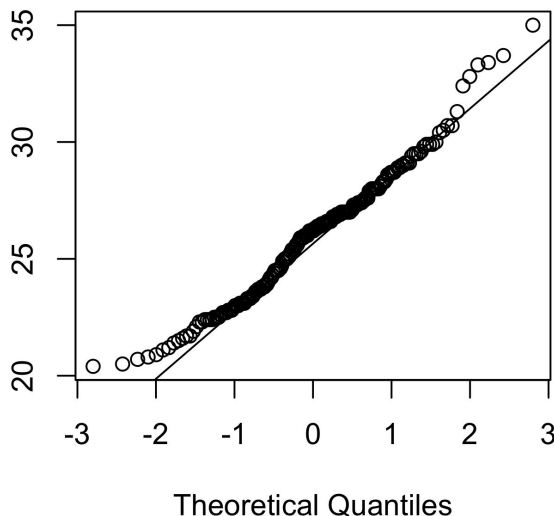
Normal Q-Q Plot



Men

```
qqnorm(women_rural_2017$bmi)  
qqline(women_rural_2017$bmi)
```

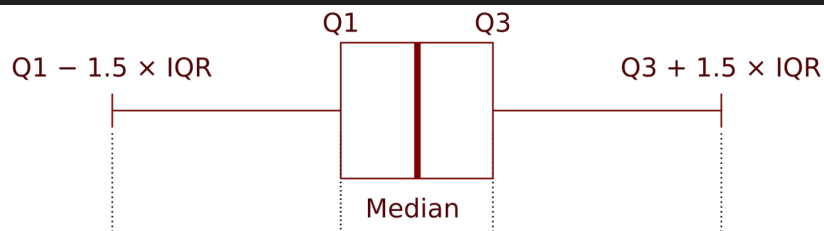
Normal Q-Q Plot



Women

- We can test the normality assumption by plotting the sample quantiles against the expected “theoretical” quantiles of a standard normal in a QQ plot
- QQ plot is the plot of the quantiles of the first dataset against the quantiles on the second dataset
- What is a Quantile? Fraction of points below the given value
- For example, 30% quantile value represents that 30% of the values in dataset lie below this point (and 70% lie above)

Visualizing difference of means



A box plot of the data set can be generated by calculating five relevant values of this data set: minimum, maximum, median (Q_2), first quartile (Q_1), and third quartile (Q_3).

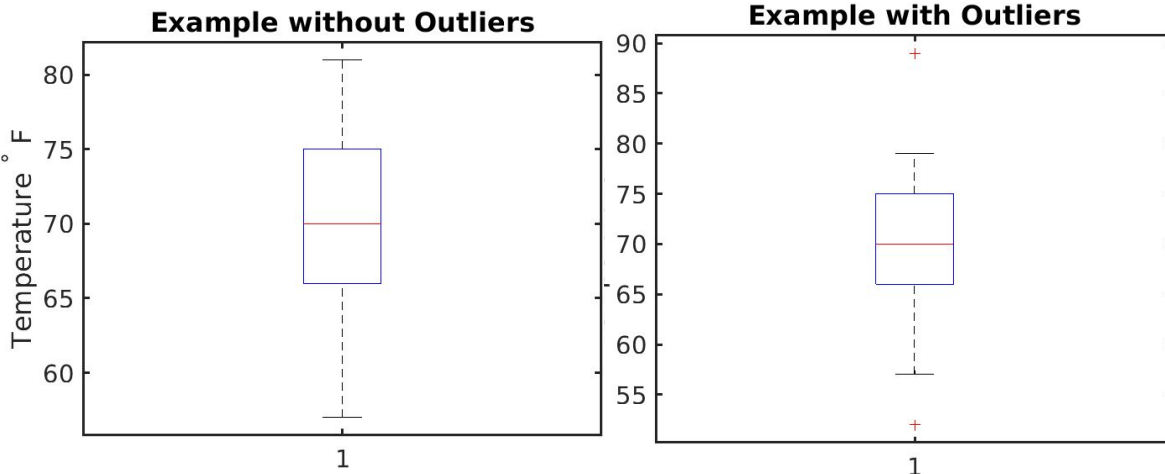
Q_1 = Quantity such that exactly 25% entries are less than equal to this quantity

Q_2 = Median

Q_3 = Quantity such that exactly 75% entries are less than equal to this quantity

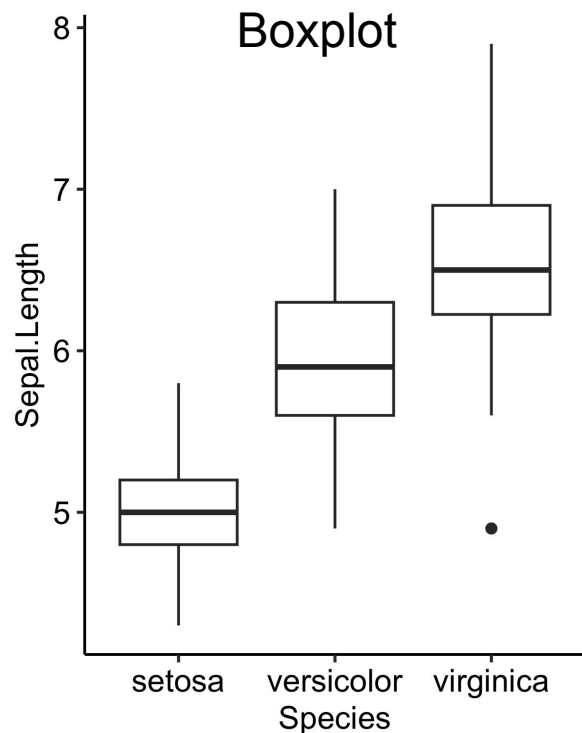
$IQR = Q_3 - Q_1$

Points beyond $Q_3 + 1.5IQR$ or $Q_1 - 1.5IQR$ are outliers

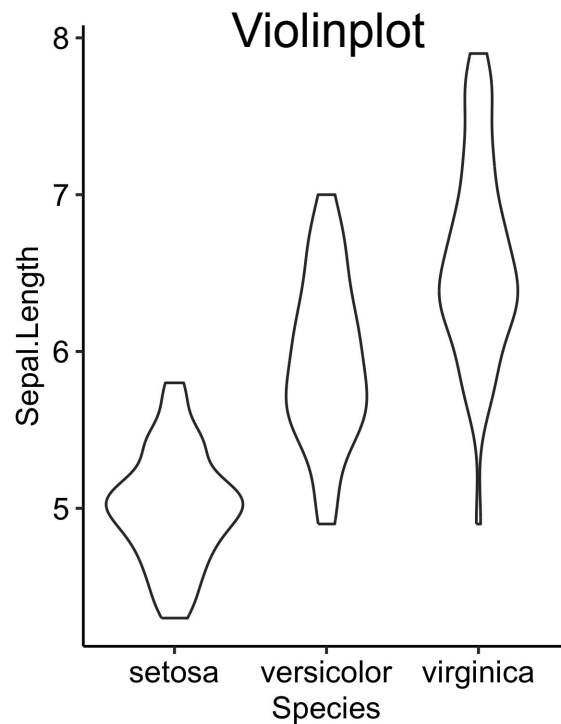


What's so special about 1.5?

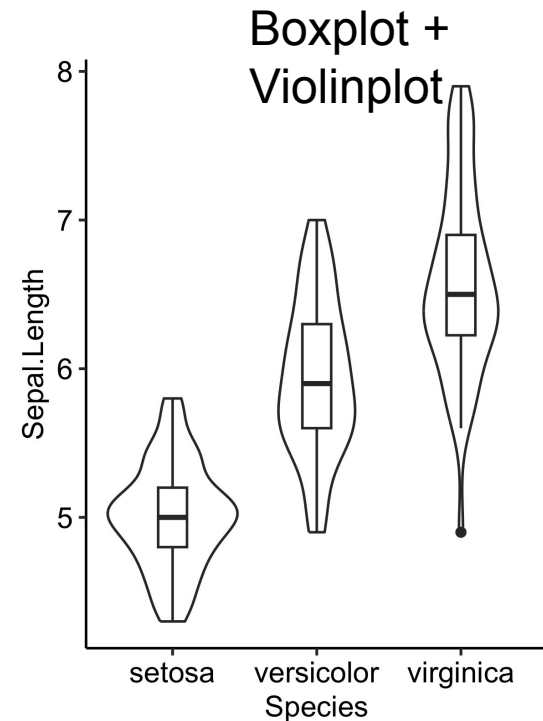
Visualizing difference of means - Boxplots and Violin plots



```
ggplot(iris,  
  aes(Species, Sepal.Length)) + geom_boxplot()  
#  
ggplot(iris,  
  aes(Species, Sepal.Length)) +  
  geom_violin()
```



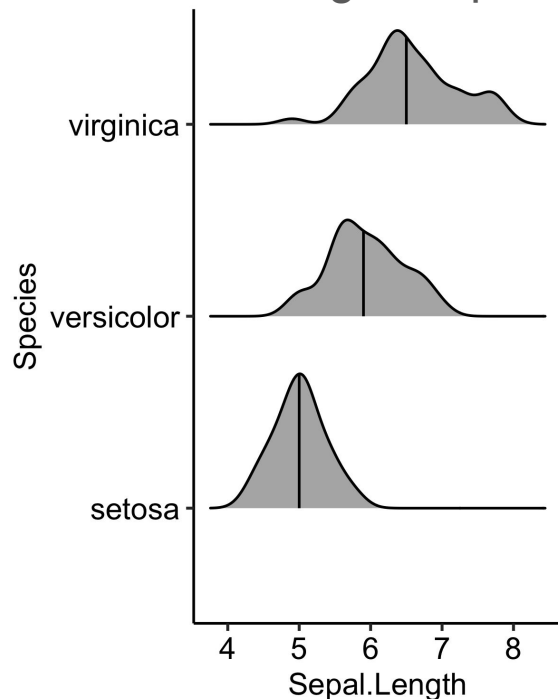
```
ggplot(iris,  
  aes(Species, Sepal.Length)) +  
  geom_violin()
```



```
ggplot(iris,  
  aes(Species, Sepal.Length)) +  
  geom_violin() +  
  geom_boxplot(width=0.2)
```

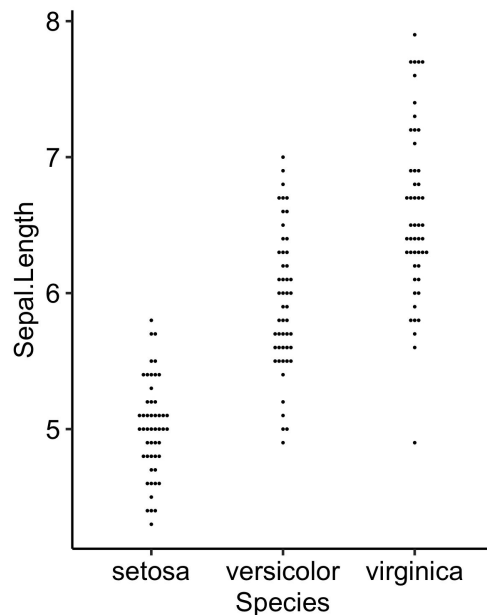
Visualizing difference of means - Bringing in more granularity

Ridgeline plot



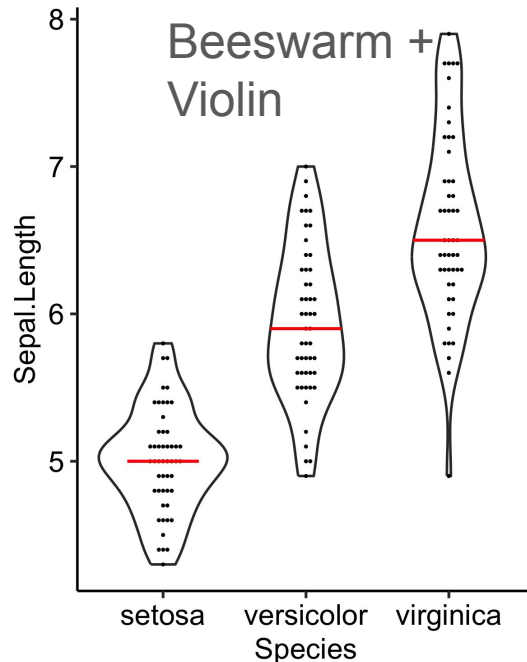
```
ggplot(iris, aes(x = Sepal.Length, y = Species)) +
  ggribges::stat_density_ridges(quantile_lines = T,
                                quantiles = 2,
                                scale = 0.7)
```

Beeswarm plot



```
ggplot(iris, aes(Species,
                  Sepal.Length)) +
  ggbeeswarm::geom_beeswarm(size =
    0.1)
```

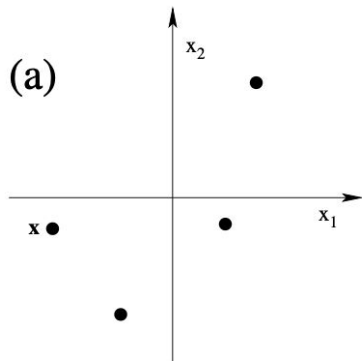
Beeswarm +
Violin



```
ggplot(iris, aes(Species, Sepal.Length)) +
  geom_violin() +
  geom_beeswarm(size = 0.1) +
  ggbeeswarm::geom_beeswarm(size = 0.1) +
  stat_summary(
    geom = "crossbar", color = "red",
    fun = median, fun.min = median,
    fun.max = median,
    fatten = 1.2, width = 0.5)
```

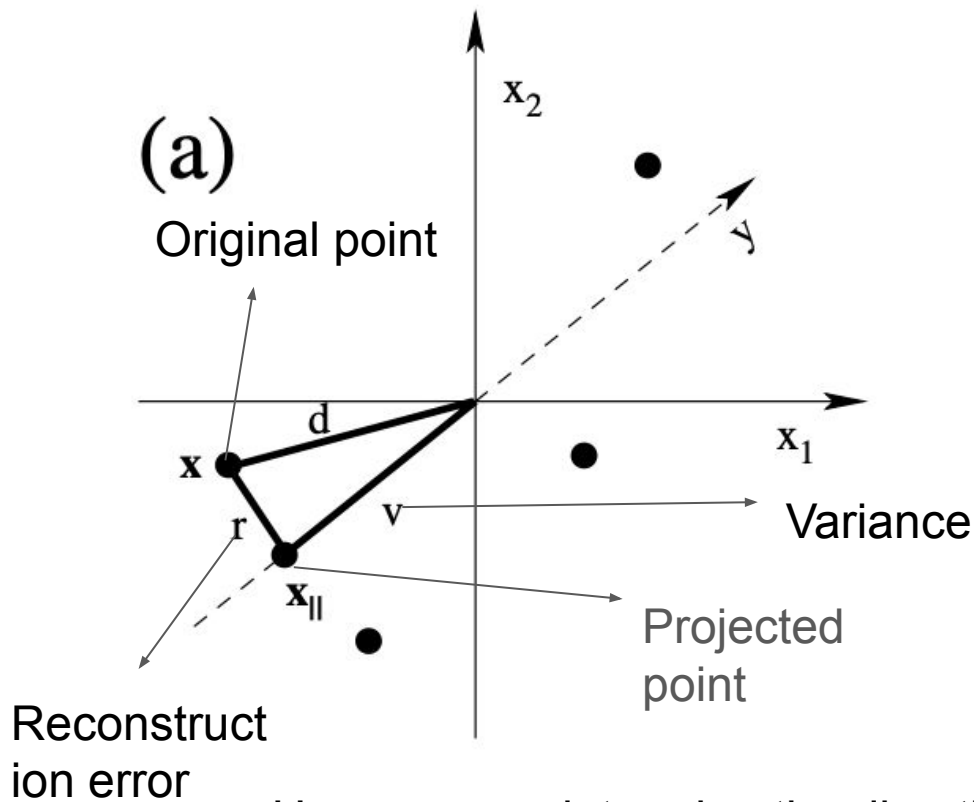
1D Principal Component Analysis (PCA)

Goal: Reduce the dimensionality of data by projecting the data into lower dimensional space such that the reconstruction error is minimized



data points in 2D

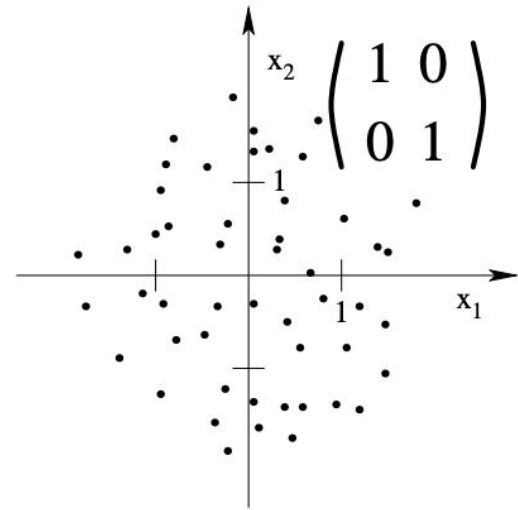
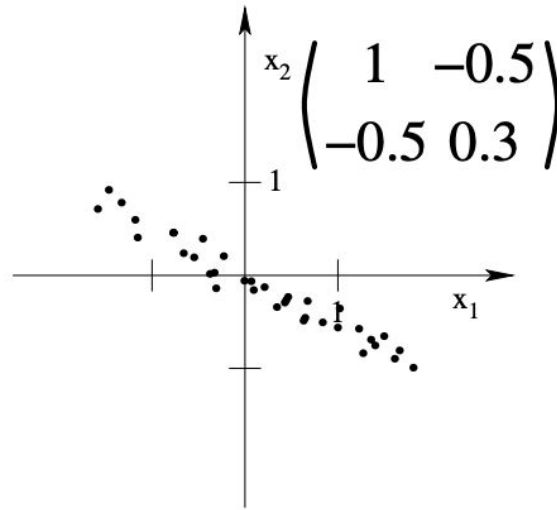
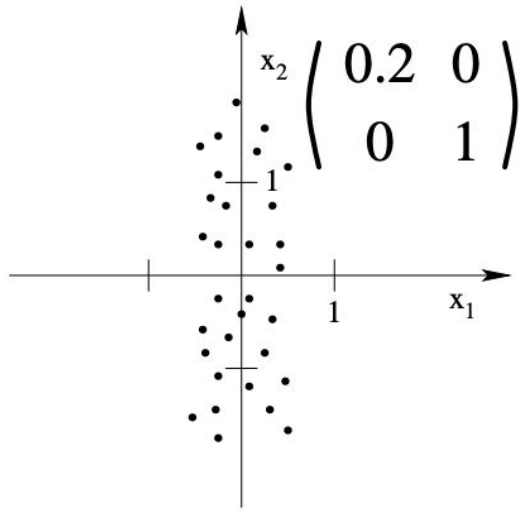
Minimizing reconstruction error = Maximizing variance



How can we determine the direction of maximal variance?

What does covariance matrix tell you about the data?

$$K_{X_i X_j} = \text{cov}[X_i, X_j] = \mathbb{E}[(X_i - \mathbb{E}[X_i])(X_j - \mathbb{E}[X_j])]$$

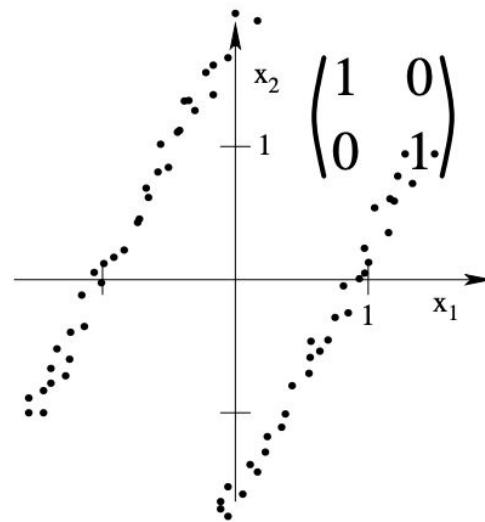
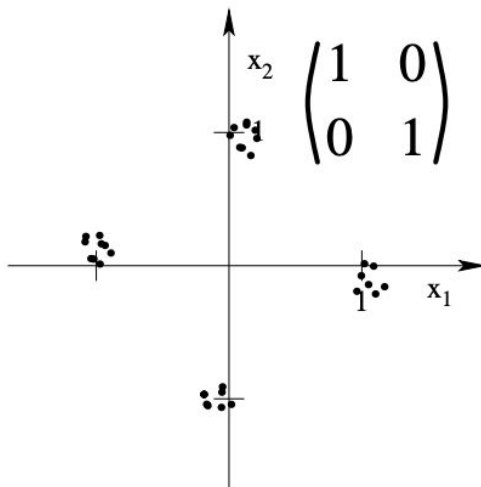
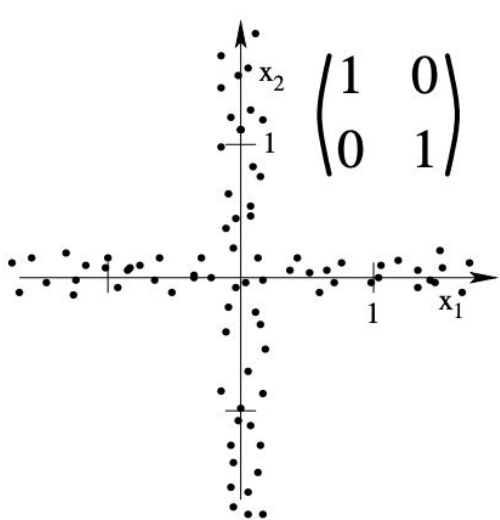


Data aligned with axes and covariance is diagonal

Data oblique wrt axes and covariance is diagonal

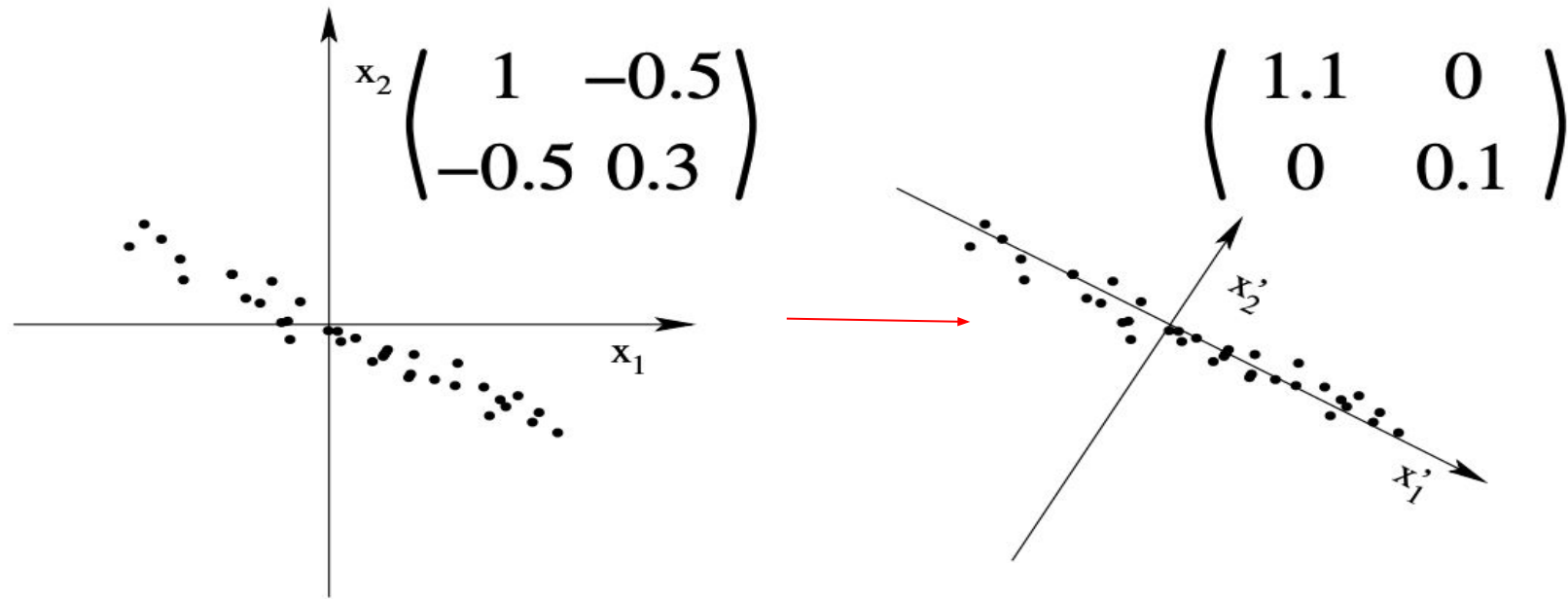
Gaussian cloud

Covariance matrix captures the general extent of data

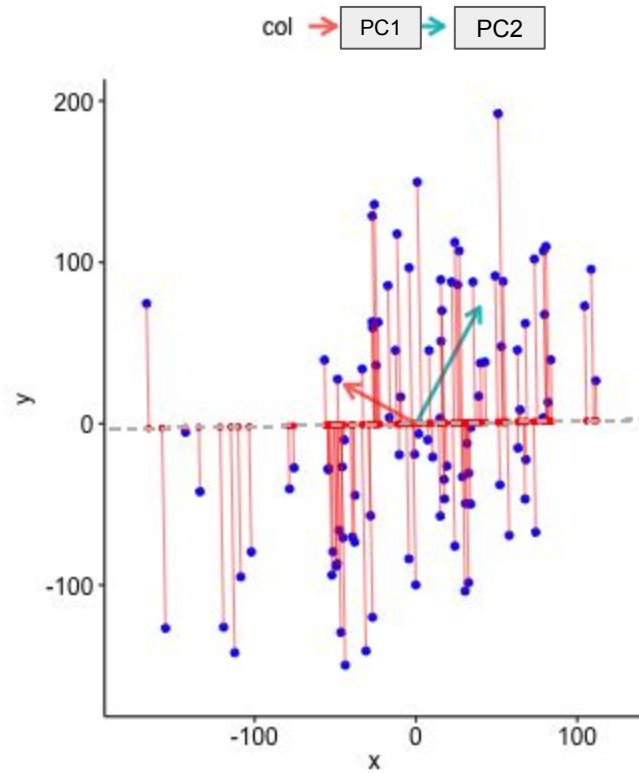


Different distributions with same covariance matrix

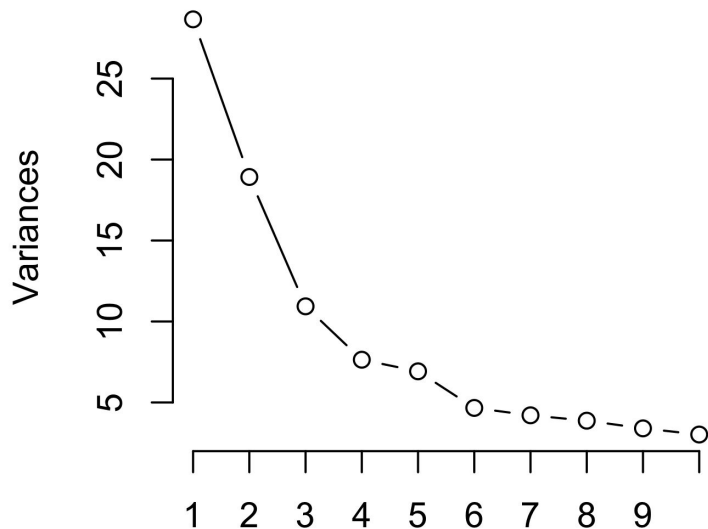
PCA rotates the axes to diagonalize the covariance matrix



PCA: the optimization



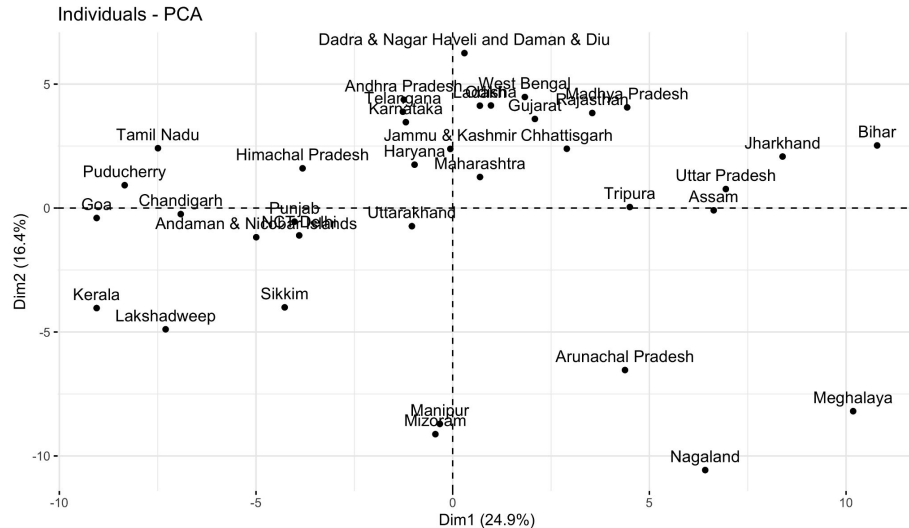
NFHS5 PCA reveals spatial clusters



Variance explained by PC
= Eigen value λ_i

Percentage variance
explained = $100 * \lambda_i / \sum \lambda_i$

```
pca <- prcomp(df, scale = TRUE)
df.pca <- as.data.frame(pca$x)
df.pca$state <- rownames(df.total.long.noindia.nona)
df.pca <- df.pca %>% select(state, PC1, PC2)
ggplot(df.pca, mapping = aes(x = PC1, y = PC2)) +
  geom_point() +
  geom_text_repel(aes(label = state), hjust = 0, vjust = 0)
# For plotting, alternatively
factoextra::fviz_pca_ind(pca)
```



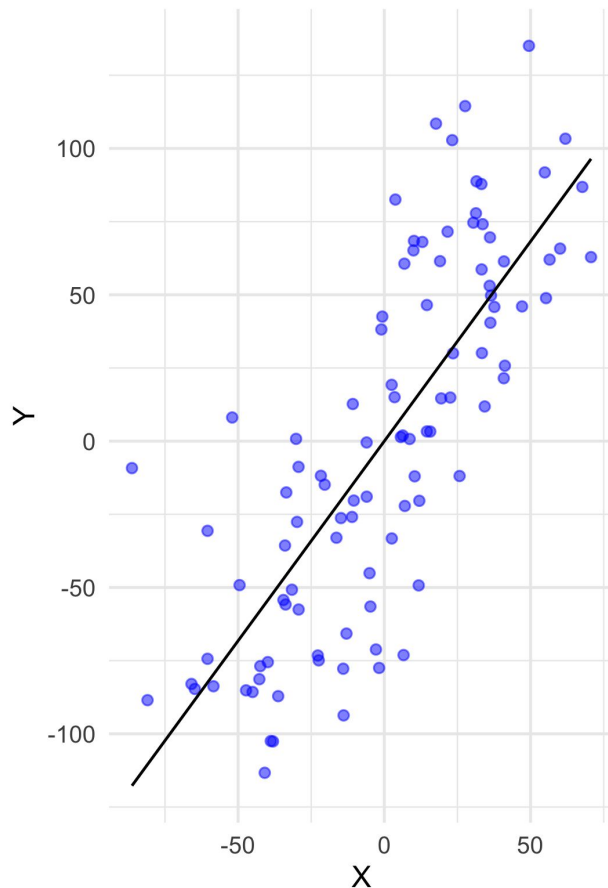
Linear regression

Regression analysis estimates the relationship between a dependent and one or more independent variable

Simplest form of regression analysis is **linear regression**: relationship between the parameters estimating the relationship between independent and dependent variable is linear

For a model to be linear, the parameters should be representable as a linear combination of the observables:

$$\hat{\beta} = \sum_i w_i y_i \text{ where the weights } w_i \text{ are estimated.}$$



Linear regression

